1 ECE 188 Notes

1.1 Reactive Component

IV relation

$$V_L = L\frac{dI}{dt} \qquad I_C = C\frac{dV}{dt}$$

Energy, using $P = IV = V^2/R = I^2R$

$$E_C = \int C\frac{dV}{dt}V dt = C\int V dV \qquad E_I = \int L\frac{dI}{dt}I dt = L\int I dI$$

1.2 Power Through Surface

Using Fourier Series

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega t - \phi_n)$$
$$i(t) = I_0 + \sum_{n=1}^{\infty} I_n \cos(m\omega t - \theta_n)$$

Energy Transferred

$$E_{cycle} = \int_0^T v(t)i(t)dt \qquad P_{av} = \frac{E}{T}$$

Orthogonality: cross-product integral terms = 0

$$P_{av} = \int_0^T V_n \cos(n\omega t - \phi_n) I_n \cos(m\omega t - \phi_m) dt \Rightarrow \begin{cases} 0 & \text{if } n \neq m \\ \frac{1}{2}V_n I_n \cos(\phi_n - \theta_n) & \text{if } n = m \end{cases}$$

Average power $< P >$

$$P_{av} = V_0 I_0 + \frac{1}{2}\sum_{n=1}^{\infty} V_n I_n \cos(\phi_n - \theta_n)$$

Root Mean Square

$$v_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt} = \sqrt{V_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} V_n^2}$$

Power Factor(PF) measures efficiency, For NonLinear Load, PF = Distortion Factor (DF) * Displacement Factor (DPF). ϕ is the PF angle, Capacitive in $\phi < 0$ (Leading), Inductive if $\phi > 0$

$$P_{rms} = v_{rms} i_{rms} \qquad PF = \frac{< P >}{v_{rms} i_{rms}} = \cos(\phi)$$

Distortion Factor and DPF, $DPF = \cos(\phi_1 - \theta_1)$

$$DF = \frac{\text{rms fundamental current}}{\text{rms current}} = \frac{I_1/\sqrt{2}}{\text{rms current}}$$

Total Harmonic Distortion (THD)

$$THD = \frac{1}{I_1}\sqrt{\sum_{n=2}^{\infty} I_n^2} \quad DF = \left[1 + (THD)^2\right]^{-1/2}$$

1.3 Complex Power Equation

Vector Power S has Apparent Power $|S|$, Average Power $Re(S)$ and Reactive Power $Im(S)$, And that $\cos(\phi) = PF$.

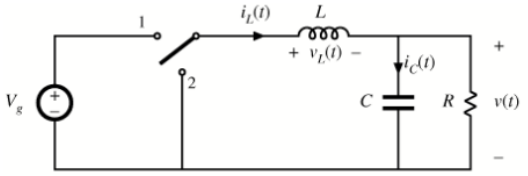
$$S = P + jQ \qquad Re(S) = P = |S|\cos(\phi)$$
$$|S| = V_{rms} I_{rms} \qquad Im(S) = Q = |S|\cos(\phi)$$

1.4 Basic Converters

Basic Idea: The average inductor current and capacitor voltage is zero in steady state, perform KVL and KCL under different state then use the Volt-Second balance on inductor and Charge-Second Balance on capacitors.

$$< V_L > = DV_L(0 - DT) + DV_L(DT - T) = 0$$
$$< I_C > = DI_C(0 - DT) + DI_C(DT - T) = 0$$

1.4.1 Buck Converter



Controlled by a switch with duty cycle D ($0 \leq D \leq 1$) in period T. Ideally, output voltage's DC component is V with a periodic ripple. Ratio M

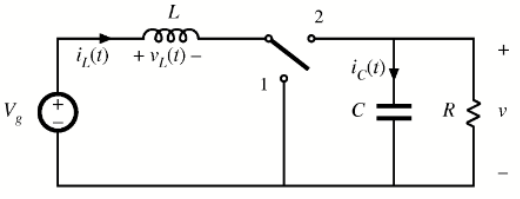
$$V_{out} = V_g \times D \qquad M = D$$

We then estimate the slope of inductor current ripple (going up, going down) to be

$$\frac{dI_L}{dt} \Rightarrow \quad \frac{2\Delta I_L}{DTs} \approx \frac{V_g - V}{L} \qquad \frac{2\Delta I_L}{(1-D)Ts} \approx \frac{-V}{L}$$

The charge q in cap is the area of triangle between ripple and DC voltage $q = \frac{1}{2}\Delta i_L(T_s/2)$

1.4.2 Boost Converter



Built by interchanging power source and load RC in buck converter. $D' = 1 - D$

$$DV_{out} = V_{in} \qquad V = \frac{V_g}{D'} \qquad I = \frac{V}{(D')R}$$

Switch in position 1, charges the inductor

$$V_L = V_g \qquad I_C = -V/R$$

In position 2, Inductor Voltage and Capacitor Current

$$V_L = V_g - V \qquad I_C = I_L - V/R$$

We then estimate the slope of inductor current ripple (going up, going down) to be

$$\frac{dI_L}{dt} \Rightarrow \quad \frac{2\Delta I_L}{DTs} \approx \frac{V_g}{L} \qquad \frac{2\Delta I_L}{(1-D)Ts} \approx \frac{V_g - V}{L}$$

1.4.3 Ripple

Utilize the fact that

$$I_C = C\frac{dV}{dt} = C\frac{2\Delta V}{dt} \qquad V_L = L\frac{dI}{dt} = L\frac{2\Delta I}{dt}$$

The Δ are the ripples. DCM occurs when the ripple is so large that the diode current drops below its DC value. $I_D \leq \Delta I_D$

1.4.4 Equivalent Circuit

With the final equations written out, draw an equivalent circuit with a transformer 1-to-N ratio.

$$P_{in} = I_g V_g = I_L V_L \qquad I_L = \frac{1}{N} I_g \qquad V_L = N V_g$$

1.4.5 Switching Loss

Replace MOS with R_{on} and Replace Diode with V_{th}, with matched polarity (pos to pos, neg to neg) Repeat analysis.

1.4.6 General Converter Ratios

Ratio $M(D) = V/V_g$

1. Ratio $M(D) = V/V_g$
2. Buck $M(D) = D$
3. Boost $M(D) = 1/D'$
4. Buck-Boost $M(D) = -D/D'$
5. None-Inverting Buck-Boost $M(D) = D/D'$
6. Bridge $M(D) = 2D - 1$
7. Cuk $M(D) = -D/D'$
8. SEPIC $M(D) = D/D'$

1.4.7 Method for Converters

Perform KVL and KCL analysis under different cycles (MOS and Diode are assumed to be in a topology cleverly designed so they will never be on at the same time). Apply V-s balance on the inductor and Q-s balance analysis on the capacitors (May be simplified by solving for equations that are always true in either cycle).

1.4.8 DCM

Ripple so large such that the assumption of operation state (MOS/Diode Either/Or) will be violated (Diode current drops to zero)

1.5 Transformer Design

High Frequency Skin Effect (in cm)

$$\delta = \sqrt{\frac{\rho}{\pi \mu f}} \qquad \delta = \frac{7.5}{\sqrt{f}} \text{ (100°C Cu)}$$

Transformer have the following design parameters

1. Transformer Design Parameters
   (a) $\rho\ \Omega cm$ Wire Resistivity
   (b) $I_{max}\ A$ Peak winding current
   (c) $L\ H$ Inductance
   (d) $R\ \Omega$ Winding Resistance
   (e) K_u Winding fill factor
   (f) $B_max\ T$ Max flux density
2. Core Dimensions
   (a) $A_c\ cm^2$ Core cross-sectional Area
   (b) $W_A\ cm^2$ Core window Area
   (c) $MLT\ cm$ Mean Length Per Turn

We can first determine L and R needed from ripple and power requirements and then determine core size (in cm^5) and air gap size (in m)

$$K_g \geq \frac{\rho L^2 I_{max}^2}{B_{max}^2 R K_u} 10^8 \qquad l_g = \frac{\mu_0 L I_{max}^2}{B_{max}^2 A_c} 10^4$$

Instead of specifying air gap length the equivalent quantity A_L(in mH/1000 turn) can be used

$$A_L = \frac{10 B_{max}^2 A_c^2}{L I_{max}^2}$$

Determine the number of turns and choose a wire size A_W (in cm^2)

$$n = \frac{L I_{max}}{B_{max} A_c} 10^4 \qquad A_W \leq \frac{K_u W A}{n}$$

Wire resistance can be computed (ρ/A_W may be provided in the datasheet for a specific type of wire)

$$R = \frac{\rho L}{A} = \frac{\rho}{A_W} n(MLT)$$

1.6 Mutual Inductance

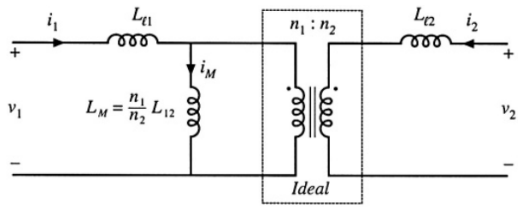
For Ideal Transformer with ratio n1-to-n2 (convention is that current enters the dot from one side exits the dot from the other)

$$v_2 = \frac{n_2}{n_1} v_1 \qquad I_2 = \frac{n_1}{n_2} v_2$$

If ideal multi-winding

$$\frac{v_1}{n_1} = \frac{v_2}{n_2} = \cdots = \frac{v_n}{n_n} \qquad 0 = n_1 i_1 + n_2 i_2 \cdots$$

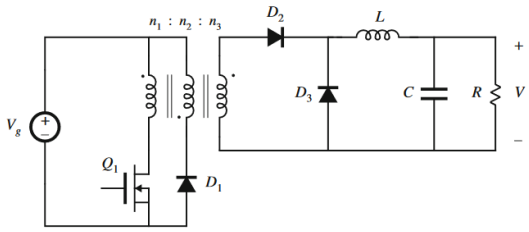
Transformer modeled by



$$\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \begin{bmatrix} L_{11} & L12 \\ L_{12} & L22 \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1(t) \\ i_2(t) \end{bmatrix}$$

1.7 Isolated Converters

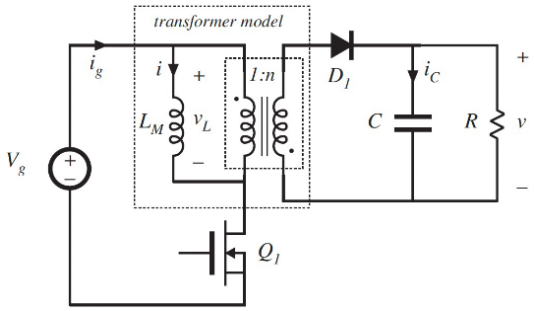
1.7.1 Forward Converter



Based on buck, but transformer with N-secondary to N-primary used as isolation.

$$V_{out} = D \frac{N_s}{N_p} V_{in}$$

1.7.2 Flyback Transformer



Based on Buck-Boost converter with N-secondary to N-primary used as isolation.

$$V_{out} = \frac{N_s}{N_p} \frac{D}{D'} V_{in}$$